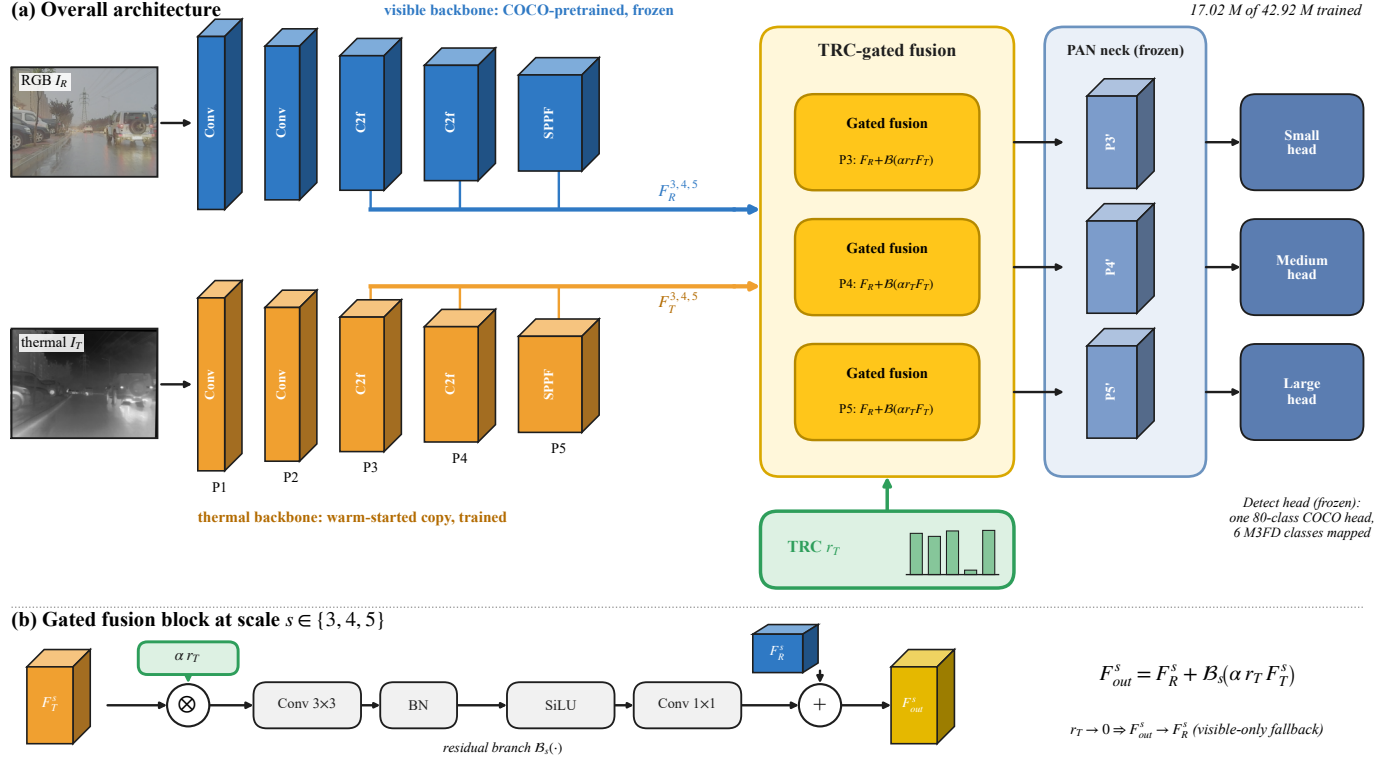

(a) Overall architecture
visible backbone: COCO-pretrained, frozen
17.02 M of 42.92 M trained
RGB I_R
Conv
Conv
C2f
C2f
SPPF
$F_R^{3,4,5}$
thermal I_T
Conv
Conv
C2f
C2f
SPPF
$F_T^{3,4,5}$
P1
P2
P3
P4
P5
thermal backbone: warm-started copy, trained
TRC-gated fusion
Gated fusion
P3: $F_R+B(\alpha r_T F_T)$
Gated fusion
P4: $F_R+B(\alpha r_T F_T)$
Gated fusion
P5: $F_R+B(\alpha r_T F_T)$
PAN neck (frozen)
P3'
P4'
P5'
Small head
Medium head
Large head
Detect head (frozen):
one 80-class COCO head,
6 M3FD classes mapped
TRC r_T
(b) Gated fusion block at scale $s \in \{3, 4, 5\}$
F_T^s
$\alpha\, r_T$
Conv 3×3
BN
SiLU
Conv 1×1
F_R^s
F_{out}^s
residual branch $B_s(\cdot)$
$F_{out}^s = F_R^s + \mathcal{B}_s(\alpha\, r_T\, F_T^s)$
$r_T \to 0 \Rightarrow F_{out}^s \to F_R^s$ (visible-only fallback)